Aerospace Readiness Mini-Challenge

Unit 0 Project Brief

Engineering Scenario

Before teams begin the major rover systems work, each student must demonstrate readiness with safety, documentation, CAD refreshers, and basic systems thinking.

Design Goal

Complete certifications and produce a small support artifact that proves you can document, fabricate, test, and reflect like an engineering team member.

Criteria	• Evidence is student-facing and organized in the engineering notebook or packet. • Prototype or support artifact connects to a future rover or mission-support need. • Documentation includes problem, criteria, constraints, sketch, build evidence, and reflection.
Constraints	• Use only approved classroom tools and materials. • Follow FabLab safety rules and teacher approval requirements. • The artifact must be small enough to finish within the Unit 0 timeline.
Required Evidence	• Certification readiness notes • Mini-challenge sketch or CAD evidence • Build/test photo or observation notes • Short reflection on readiness for POE projects
Checkpoints	• Identify readiness expectations • Complete tool/safety evidence • Build or document the mini-challenge artifact • Submit final reflection

Final Design Review Expectations

Your final review should clearly connect the problem, evidence, prototype decisions, testing results, limitations, and next recommended improvement. Strong teams show how data changed the design rather than only describing what they built.

Student Deliverable Reminder

Keep all sketches, calculations, data tables, photos, revision notes, and decisions in your engineering notebook or assigned project packet. The notebook should make your design process visible.